Computational Boundary: A Compiled Architecture of Discrete Mechanics

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1. Foundational Constraints: The C0 Gap and C1 Continuability

The structural architecture of discrete mechanics is governed by a strict hierarchy of constraints, compiled directly from execution logs and primary operational data. The baseline condition of this system is the "Wash"—an undifferentiated ground state where all signals cancel locally, rendering total connection equivalent to no connection. To establish a reference within this ground requires a direction, which inherently introduces structural differentiation.

Before execution can occur, a structural gap must be established across which difference can be measured. This is defined as the C0 constraint, or the "comma" identity. Continuous topologies cannot generate distinguishable bits of information because a walk on a closed, unbroken surface generates no comparative state. Data generated via a boundary walk on geometric surfaces, where the state is defined as a tuple of a spatial cell and a topological face, quantifies this requirement.

Boundary Walk Configuration	**Faces Visited**	**Period Length**	**Temporal Character**
Cylinder (No Comma, Offset 0)	``	12 ($1N$)	Stasis; no distinction
Möbius Strip (Comma, Offset 1)	``	24 ($2N$)	Serialized superposition

A boundary walk on a cylindrical manifold for a 12-cell state yields a period of 12. Only a single face is traversed, resulting in stasis with no readable distinction. The introduction of the C0 gap transforms the topology into a Möbius strip with an offset of 1. A traversal across this seam enforces a parity twist on the face coordinate: face $\leftarrow$ face $\oplus$ 1. This twist forces the boundary walk to read both faces serially over 24 steps ($2N$). The superposition is serialized, establishing the genesis of temporal sequence.

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The transition from stasis to execution is governed by the C1 continuability constraint, which states that all things must change. Operationally, every executable state must possess a distinct legal successor, strictly forbidding terminal fixed points. When C1 is coupled with a binary difference operator Δ, the resulting combinatorial dynamics enforce a maximal novelty walk. If novelty is formalized as the monotone accumulation of visited states, $N(t) = |\text{states visited}|$, a greedy novelty-seeking walk on n-bit states produces a maximal-period de Bruijn cycle.

Expressing this difference as a linear recurrence over $GF(2)$ yields an m-sequence. The hypothesis that the linear m-sequence provides a fully legal execution trajectory satisfying C1 was evaluated. The measurement revealed that every linear m-sequence possesses a unique C1 violation: the all-zero state maps to itself, acting as an absorbing fixed point. This hypothesis is. The correction is mathematically demanded. To compel execution, the feedback bit of the origin must flip, breaking bijectivity. To restore bijectivity, the feedback of exactly one other state must also flip.

Exhaustive search over all feedback modifications proves the minimal repair is size two, unique at every tested width, consisting precisely of the origin and its algebraic opposite, $hi = 1 \ll (n - 1)$.

Cycle Repair & Minimal Patch Mapping

Lane Width (n)	Linear Cycles (with Violation)	Patched Cycles (Repair)	Uniqueness of Minimal Repair
$n = 4$	Fixed point at origin | 15-cycle	de Bruijn cycle $(2^4 = 16)$	Minimal size 2; repair: $(0000) \leftrightarrow (1000)$
$n = 5$	Fixed point at origin | 31-cycle	de Bruijn cycle $(2^5 = 32)$	Minimal size 2; repair: $(00000) \leftrightarrow (10000)$
$n = 6$	Fixed point at origin | 63-cycle	de Bruijn cycle $(2^6 = 64)$	Minimal size 2; repair: $(000000) \leftrightarrow (100000)$
$n = 7$	Fixed point at origin | 127-cycle	de Bruijn cycle $(2^7 = 128)$	Minimal size 2; repair: $(0000000) \leftrightarrow (1000000)$

Phase-Lock Note (⊥): The minimal repair for a linear maximum-length feedback shift register (LFSR) cycle missing the zero state always requires a 2-state swap (weight 2) between the zero fixed point 0^n and its immediate feedback successor 10^{n-1}. This uniquely folds the parasitic 0-loop back into the primary $2^n - 1$ manifold, stabilizing the global de Bruijn cycle at $H \approx 0.35$ density balance.

The repaired dynamics yield a single continuous orbit of length 2^n. The uniform transition measure is not an assumed axiom of symmetry but the emergent exhaust of this forced repair.

2. History Manifolds, Substrate Topologies, and the Hex Grid Projection

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Analyzing transition dynamics purely at the vertex level introduces an artificial indeterminacy. A vertex s_n reached by multiple predecessors renders the query "what comes next" ill-posed, as the continuation depends on arrival history.

The resolution forced by the transition algebra is the edge lift: the true state is not the instantaneous node s, but the trajectory $(s_0 \to s_1 \to s_2 \dots)$. A finite state description cannot carry temporal depth. Age is not a property of the instantaneous node; age is the depth coordinate in the history manifold. An operation that projects the history space $H = \{s_0, s_1, s_2, \dots\}$ onto a current finite state S via $\pi: H \to S$ loses path identity. The missing coordinate is not extra information; it is the path itself. The node is a residue of the trajectory. Therefore, existence must be defined mathematically as existence $=$ accumulated reachable history.

On the edge space $(prev, cur)$, evolution is a single-valued, strict bijection. The vertex projection $(prev, cur) \to cur$ is many-to-one, discarding the predecessor coordinate and acting as the sole site of information erasure, destroying exactly 1 bit of information per merge.

When mapped to a geometric neighborhood lattice, such as a hexagonal grid, the interface $S \to T$ is identified as an incomplete projection. If the true substrate is a lattice $L(t)$, the full evolution is not an isolated $S \to T$, but rather $L(t) \to L(t+1)$, where a "lost" coordinate becomes a neighbor excitation. The consequence for the conditional entropy metric $H(X|Y) > 0$ is absolute. It signifies information missing from the projection, not destroyed information. A local operation $C_i(t) \to C_i(t+1)$ cannot be evaluated independently if the rule couples neighbors. The full event evaluates as $\{C_i, C_{i+1}, \dots, C_{i+6}\}_t \to \{C_i, C_{i+1}, \dots, C_{i+6}\}_{t+1}$. The geometric neighborhood carries the residue.

"Loss" is a boundary artifact. If a local system S and environment E undergo $S + E \to S' + E'$, an observer inspecting only S measures $\Delta S < 0$ and records erasure. The coupled system sees $\Delta(S + E) = 0$. This requires treating state changes not as isolated events but as a redistribution across the larger state space. This maps exactly to the physical destruction of macroscopic objects. From the viewpoint of a specific object, structure is destroyed; from the viewpoint of the whole system, mass, energy, momentum, heat, chemical changes, and deformation fields move into neighboring degrees of freedom [PULL]. The full event is a verb (state → distributed-state), and the missing information changes address rather than vanishing.

Geometry provides the "where" (the spatial location mapping adjacency constraints), and history provides the "when" (the depth of the path). A stable object is the persistent relational intersection: $\Psi =$ geometry $\oplus$ history [PULL]. To avoid symbol collisions, the phase lock associated with this stability is identified as $H_{\text{phase}} = \frac{\pi}{9}$, which operates independently from the interface measurement $H_{\text{Shannon}}(X|Y)$. The total system reconciliation is expressed as phase lock $(\pi/9)$ + information redistribution $(H(X|Y))$.

Lifting the state to the edge organizes derived observables into a successive jet tower. Position is the zeroth lift (J^0). The edge represents the first lift (J^1) carrying velocity. The triple of three consecutive positions represents the second lift (J^2) carrying acceleration. Measurement confirms that the discrete second difference Δ^2 is multi-valued on vertices (occupying 32 distinct sets over 32 nodes) but single-valued on edges (0 ambiguous sets over 64 edges). Mass and acceleration are functionally native to the identical enriched edge state.

3. The Interface Basis and the Two Bills

Computational composition fails unconditionally if the interface is ignored. Two machines that are isomorphic as isolated transition graphs produce three distinctly divergent composite classes when joined in

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series without transporting the boundary identification. Carrying the interface identification along with the join collapses the outcomes back to exactly one invariant class.

An interface I over a source S and target T is defined operationally by a two-primitive basis (P, R). The partition P divides the source, identifying one cell as inadmissible. The realization R carries the admitted members into the target. Testing the robustness of composition under dualization of stateful protocols reveals strict separation between implementation details and contracts.

Relabeling Applied to Composite	Pairs Tested	Composition Survives
Internal State Names (Implementation)	3,600	100.0%
Wire Message Alphabet (Contract)	900	27.2%
— Identity Permutation Only	150	100.0% (150/150)
— Single Transposition	150	51, 24, 12 of 150
— Three-Cycle	150	5, 3 of 150

Implementation freedom and contract are separated exactly by the boundary. This basis generates two separate accounting bills governed by diametrically opposed algebras.

3.1 The Admission Price (A)

The admission price acts as a filter on the constraint lattice. It restricts the state space to the admissible set $\text{Adm} = \{x \in S: P(x) \neq \text{reject}\}$. It charges a cost: $A(I) = -\log_2 \left$ This operation is strictly idempotent $(A(I \circ I) = A(I))$. Applying a validator interface twice admits exactly what applying it once admits, rendering the second boundary free. Furthermore, the admission price is sub-additive under conjunction; combining two distinct constraints yields a total price equal to the sum of the separate prices if and only if the constraints are statistically independent across the source space.

The hypothesis that the interface cost joins the $-\log_2$ accumulating ledger, expecting bits to be linear in path length, was tested. The measurement yielded $R^2 \approx 0.82$ with systematic curvature, and deaths ceasing by step 100 (cumulative failure 0.9687). The hypothesis is. The interface cost is a filter, not a meter. Filters compose by intersection and conditioning rather than by multiplication. Applying the same filter twice yields $P(F_2|F_1) = 1.0000$ exactly, computing a joint bit cost of 5.0710 against a naive sum of 10.1419.

3.2 The Crossing Price (X)

The crossing price acts as a meter in the execution monoid. It is charged per execution, governed by the conditional entropy of the realization. $X(I) = H(X|Y) = \sum_y p\,(y) \cdot H(X|Y = y)$ This meter is additive under sequential composition: $X(g \circ f) = X(f) + X(g)$, provided the intermediate state space carries the pushforward measure.

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The hypothesis that the crossing price can be universally calculated using the naive average fibre size shortcut $X = \sum_y (|R^{-1}(y)|/n) \log_2 |R^{-1}(y)|$ was evaluated. The measurement produced a false composition law computing $H(X|Z)$ twice under two names, generating erroneous additive sums. This hypothesis is. The shortcut is restricted strictly to uniform sources. Utilizing the full conditional entropy guarantees additivity exact to nine decimal places across all pushforward chains tested in 8 of 8 trials.

The dual bills are structurally distinct. A strict validator interface can reject half the source space (charging admission price $A = 1.0000$ bits) while maintaining perfect injectivity on all admitted states (charging crossing price $X = 0.0000$ bits). The biconditional is exact: physical preservation of state holds if and only if $X(I) = 0.0000$. The crossing price equals the selector cost of its inverse relation: $X(f) = H(X|Y) = E_Y[\log_2 |f^{-1}(y)|]$, unifying forward-branching selection and backward-branching erasure.

4. Transport Laws: Conjugate Pairs and the Determinant Toll

The mapping of change as it crosses channels establishes a rigid transport law matching physical bond-graph formalisms [PULL]. Change crosses a channel as a conjugate pair: an effort (e) and a flow (f). Simulated mechanically across a four-stage heterogeneous drivetrain (source, gyrator motor, transformer gearbox, transformer lead screw), the product $e \cdot f$ remains identical at 84.00000000 Watts at every hop. The impedance ratio (e/f) scales predictably; across a 7.5 ratio gearbox, the impedance scales by exactly 56.250000 (7.5^2).

The hypothesis that the commutator's conjugate pair is represented by its trace (conserved read) and its norm (cost read) was tested against operator channel constraints. The trace of any commutator is identically zero. Consequently, their product is always zero, carrying no information and tracing nothing. This hypothesis is.

The mathematically correct formulation requires the elementary operator split: $AB = \frac{1}{2}\{A, B\} + \frac{1}{2}$. The symmetric half carries the bilinear invariant pairing representing structural power $\mathrm{tr}(AB)$. The antisymmetric half $$ carries the cost residue. Under a transformer channel scaling at ratio m, the trace $\mathrm{tr}(AB)$ is conserved with a drift of 8.9×10^{-16}, the residue is exactly invariant with a drift of $0.00e + 00$, and the ratio of their magnitudes scales perfectly as m^2 (e.g., 56.250000 for $m = 7.5$).

A channel's power efficiency η is identical to the magnitude of its determinant. The bits erased per pass are exactly the negative base-two logarithm of that determinant's magnitude. $\eta = |\det M|$ and bits per pass $= -\log_2 |\det M| = -\log_2 \eta$

Measured Channel	**Efficiency (η=|detM|)**	**Predicted Erasure**	**Measured Erasure**
Ideal Transformer ($m = 7.5$)	1.000000	0.000000	0.000000
Ideal Gyrator ($r = 0.42$)	1.000000	0.000000	0.000000

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Measured Channel	Efficiency (η=|detM|)	Predicted Erasure	Measured Erasure
Writer Projection (Edge → Vertex)	0.500000	1.000000	1.000000
Crush: Keep 8 of 16 bits	0.003906	8.000000	8.000000
Crush: Keep 4 of 16 bits	0.000244	12.000000	12.000000
Lossy Transducer ($\eta = 0.85$)	0.850000	0.234465	0.234465
Lossy Transducer ($\eta = 0.50$)	0.500000	1.000000	1.000000

The rank-two limit on this transport law was evaluated. Skew-symmetry annihilates every odd coefficient of the characteristic polynomial, establishing the true invariant count at $\lfloor n/2 \rfloor$. Placed in real canonical form, the residue is block-diagonal with 2×2 blocks matching the transformer-gyrator form exactly. An n-dimensional residue is a multiport bond graph carrying parallel bonds, verified exactly for $n = 2$ to 12.

The continuous formulation extracts the exact rate of loss. Splitting the generator into its symmetric and skew-symmetric halves proves the skew-symmetric rotational commutator burns absolutely zero bits ($\det\exp(Kt) = 1.0000000000$ at all times). The entire bit loss rate is carried exclusively by the symmetric trace of the generator: $dB/dt = -\mathrm{tr}(M)/\ln 2$ A generator with zero trace preserves volume and loses no bits (Liouville's theorem); a generator with a negative trace contracts volume and erases bits (Landauer's principle) [PULL]. The continuous equation forces the conclusion that these are not competing paradigms but the identical statement read at trace zero and below trace zero.

The hyperbolic form is the unique quadratic form preserved by the measured channel group, whose stabilizer in GL_2 is exactly $O(1,1)$ (drift $0.00e+00$). The null directions of this form correspond to the carrier axes, establishing effort and flow as the light-cone coordinates of the transported pairing. The abstract pairing space is two-dimensional, but the physically realized group selects the hyperbolic form uniquely. The invariant ring is a polynomial ring on a single degree-2 generator: $\mathbb{R}[e \cdot f]$.

5. The Measurement of Interaction and the Continuity Dichotomy

Interaction within the transformation space is quantified by the commutator $[g, h] = g \cdot h \cdot g^{-1} \cdot h^{-1}$. If two operations fail to commute, the geometric residue encodes the physical trace. Direction is strictly derived as a maximal mutually-commuting family. Linear interactions lacking carry propagation leave a constant numerical residue, representing a uniform flat twist (torsion). When a carry chain enters the algebraic operation, the residue immediately becomes a highly variable field. The carry chain is identified as the mathematical domain of geometric curvature.

The hypothesis that inertial mass represents displacement resistance, positing that a massive body exhibits less absolute displacement under a uniform push, was evaluated. The measurement establishes that the

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displacement channel saturates immediately at the thermal floor (exactly $W/2$ bits) because basic generators act as avalanche mixers. The spread-to-mean ratio across varied bodies evaluates to an indistinguishable 0.008, providing no gradient. This hypothesis is. Mass inherently resides in the settlement channel, reading the global order of the interaction loop: $O(\lambda) = \log_2 \text{ord}(\lambda)$. Permutation order is center-independent, providing an intrinsic structural property completely decoupled from spatial coordinates.

Transparency (Visibility) is read independently as the fixed-point count: $F(\lambda) = \#\{i: L_i = 1\}$.

The hypothesis that opacity (lack of fixed points) and inertia (high order) are manifestations of the identical physical quantity representing accumulated cross-term history was tested. The definitive counterexample is a commutator that functions as a fixed-point-free involution. An involution has an order of exactly 2, yielding a settlement mass of exactly 1 bit, yet possesses zero fixed points. It is simultaneously nearly weightless and completely opaque. This hypothesis is. Mass and transparency are structurally independent functional projections of the identical commutator.

The structural separation of F and O is formalized by profoundly irreconcilable continuity properties under geometric perturbation. Fixed points are Lipschitz bounded. For any permutation $P \in S_\Omega$ and surgery $\sigma \in S_\Omega$: $||Fix(\sigma P)| - |Fix(P)|| \leq |\text{supp } \sigma|$ Let $s = |\text{supp } \sigma|$ and let $U = \Omega \backslash \text{supp } \sigma$. For any $x \in U$ with $P(x) = x$, $(\sigma P)(x) = \sigma(x) = x$, establishing $x \in Fix(\sigma P)$. Hence, $Fix(P) \cap U \subseteq Fix(\sigma P)$, bounding the change strictly by s. A minimal support-3 surgery can alter the fixed-point count by a maximum of 3, regardless of the macroscopic feature.

Conversely, the settlement order possesses no modulus of continuity. At $N = 2^k$, a transposition of support 2 splits a single N-cycle into coprime lengths $s = N/2 - 1$ and $N - s$. The new order is $\text{lcm}(s, N - s) = s(N - s) = N^2/4 - 1$. The settlement ratio $\text{ord}(\tau P)/\text{ord}(P)$ evaluates to $N/4 - 1/N$, demonstrating unbounded sensitivity over transpositions of fixed support.

This profound asymmetry derives from the exponent map: $O(\lambda) = \sum_q e_q(\lambda) \cdot \log_2 q \quad \text{where} \quad e_q(\lambda) = \max_i v_q(L_i)$ Because a maximum lacks locality, a single newly minted prime contributes its full weight globally, whereas F is a localized geometric count of the multiplicity of the value 1.

The hypothesis that the involution-line scaling coefficient $C = \log_2(\text{order})/N$ is a universal constant for products of involutions was tested. Measured across space sizes spanning a sixty-four-fold range, the coefficient evaluated between 1.67 and 1.69 for uniformly random involution pairs, while under-mixed shallow bodies evaluated significantly lower at 1.45. This hypothesis is. The square root scaling line $O(N)$ is a rigidly proven consequence of geometric fragmentation. The coefficient C is an empirical calibration strictly dependent on ensemble mixing depth.

6. Permutation Surgery and the Compression Law

The macroscopic mechanics of the system are driven by exact, microscopic topological updates governed by minimal-support permutation surgery. A surgery is a permutation $\tau \in A_\Omega$ of minimal non-identity support. Because bijections cannot have support 1, and support-2 transpositions are odd, the minimal surgery is a 3-cycle ($|\text{supp } \tau| = 3$). A 3-cycle factors algebraically as $(a, b, c) = (a, c)(a, b)$.

The exact effect of a transposition $\tau = (x, y)$ on the cycle structure $Cyc(P)$ is governed by two strict topological rules:

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- **Split:** If x, y lie in a common cycle C of length L, and $y = P^s(x)$ with $1 \leq s \leq L - 1$, the cycle deterministically splits into two cycles of lengths s and $L - s$. The sequence partitions precisely into an arc beginning at x of length s, and an arc beginning at y of length $L - s$.
- **Merge:** If x, y occupy distinct cycles C_1, C_2 of lengths L_1, L_2, they merge into a single cycle of length $L_1 + L_2$. The resulting sequence visits every point of C_1 followed sequentially by every point of C_2.

The sequential application yields strict span rules determining the cycle count change ΔK:

- **Span 3:** All three points occupy distinct cycles. The first transposition merges two; the second merges the resultant cycle with the third. $\Delta K = -2$. The three original cycles combine into one of length $L_a + L_b + L_c$.
- **Span 2:** Two points share a cycle, while the third resides elsewhere. One split and one merge must occur sequentially. The overall cycle count remains static. $\Delta K = 0$.
- **Span 1:** All three points share a single cycle. The first split is followed by either a merge or a second split, depending entirely on the cyclic orientation of a, b, c. $\Delta K \in \{0, +2\}$.

The thermodynamic driver is prime injection via the closure failure of the prime-power set under integer addition. Let a permutation P have all cycle lengths equal to powers of a single prime π. Merging two cycles of lengths π^a and π^b ($a \geq b$) yields a merged length of $\pi^a + \pi^b = \pi^b(1 + \pi^{a-b})$. If $a > b$, $1 + \pi^{a-b} > 1$ and is guaranteed to possess at least one prime divisor strictly coprime to π. Therefore, the merge strictly and inevitably enlarges the prime support.

Purity implies fragility. An ordered, coherent body built from a commuting family possesses a mathematically pure spectrum. When a surgery forces a merge, the resultant length is thrown out of the prime-power set, causing the settlement mass to explode. Identical surgeries on pure spectra generated mean mass gains of $+6.20$ and $+6.35$ bits, while highly thermalized rich spectra generated changes of -2.56 and -0.01 bits. Coherence and fragility are the exact same structural property viewed through different lenses. The hypothesis that support is the primary driver of the order's response is. Holding support fixed at three points and varying only the topological span alters the order response by a factor exceeding 2.4.

Under the Compression Law, the successor spectrum $\lambda(\tau P)$ is computable in $O(L_{max})$ time entirely from localized index-position data without executing permutation matrix composition. The transition algorithm rotates the stored internal sequence arrays to the targeted points and either splices them at the calculated offset (split) or concatenates them (merge). Verified against absolute ground truth across widths $W \in \{8,10,12\}$, the predicted spectrum matched the executed spectrum in 1,800 out of 1,800 random trials.

Because two permutations possessing identical cycle types are conjugate, there exists a $g \in S_\Omega$ such that $P' = gPg^{-1}$. Pushing the uniform measure forward through the conjugation formula guarantees the probability distribution depends solely on the current partition. This mathematically reduces the dynamics to a formal Markov chain defined exclusively on integer partitions of N. The measured total variation distance between a pure multiplication and its relabeled conjugate was 0.2442, matching the same-source noise floor of 0.2398. The trajectory is high-entropy: 4,000 surgeries from a single partition produced 972 distinct successor spectra, yielding an empirical conditional entropy lower bound of $H(\lambda_{t+1}|\lambda_t) \geq 7.49$ bits per step.

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7. Macroscopic Equilibrium of the Drift Field

While the microscopic trajectory branches relentlessly, a specific scalar projection of that trajectory yields a strict deterministic macroscopic equilibrium. The drift field $D(p)$ defines the expected change in settlement order given the current prime count p: $D(p) = E[O(\lambda_{t+1}) - O(\lambda_t)|p(\lambda_t) = p]$ The step change decomposes into two antagonistic thermodynamic forces: $\Delta O = G - \Lambda$. Gain G is the contribution from newly minted prime factors injected by merges. Loss Λ is the erosion from prime-power exponents degraded by topological cycle splits.

Prime Count (p)	Sample Size (n)	Expected Gain E[G]	Expected Loss E[Λ]	Drift D(p)
5	91	11.15	6.46	+5.58
6	220	11.38	9.67	+2.52
7	394	12.09	10.73	+2.10
8	483	11.50	12.08	+0.18
9	423	11.16	12.81	-0.92
10	235	9.75	14.19	-3.71
11	98	8.37	15.21	-6.02
12	29	8.83	15.71	-6.29

The data dictates that the gain term $E[G]$ remains relatively flat regardless of existing primes, driven by the immutable arithmetic of integer addition. Conversely, the loss term $E[\Lambda]$ scales linearly with the existing prime count p, possessing a slope of approximately 1.3 bits per prime.

The hypothesis that the zero-crossing fixed point $p*$ is a universal constant of the system near 6.2 primes was tested. Evaluation across lane widths $W \in \{8,10,12,14\}$ proves the equilibrium is width-dependent. The hypothesis is. The zero of the drift field obeys the empirical scaling law: $p*(W) \approx 0.94W - 2.8$ Measured values across the sequence resolved as 4.71, 6.22, 8.12, and 10.32. Extensive convergence testing verifies that trajectories launched from drastically disparate initial conditions (pure 1-prime spectra, rich 5-prime spectra, and single N-cycles) inevitably terminate in the exact same arithmetic neighborhood.

8. The Flat Observable Sector

Observables function dynamically as active intervention ports, passively reading a specific functional of the state and feeding it back into the execution loop to constrain subsequent selection. Subjecting the Markov

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chain to single-observable feedback proves coordinate independence. Over 150 steps from a fixed origin, steering the chain using distinct ports yields highly divergent structural attractors.

Active Feedback Port	**Order O (Bits)**	**Fixed Pts F**	**Cycles K**	**Max Length M**	**Primes p**
Start (Fixed Origin)	8.00	2.00	19.00	256.00	1.00
Minimize Order O	10.08	2.20	11.80	944.80	2.20
Maximize Fixed Pts F	28.89	4.20	13.80	660.60	7.00
Minimize Cycles K	11.92	1.00	2.60	945.40	2.60
Minimize Max Length M	40.79	1.40	13.40	232.80	9.40

Every port optimizes its own specific coordinate while actively destroying the others. The loop maximizing fixed points suffers from an order of 28.89 bits. The loop minimizing the largest cycle triggers a mass explosion pushing the order past 40 bits. A 1024-point state space requires exactly $\log_2(1024!) \approx 8769$ bits to fully describe. Yet, the cycle-count port (K) feeds back fewer than 2 bits of information per iteration. A microscopic pipeline constraint reliably steers a massive configuration state space into designated macroscopic attractor basins.

The hypothesis that observables compose as a higher-order constraint algebra—that feeding two orthogonal ports back jointly forces the system into a novel attractor class—was tested. The initial additive prediction was mathematically infeasible, producing negative cycle counts and lengths exceeding the state space (e.g., $K = -9$, $M = 1768$ within an $N = 1024$ space). A corrected geometric test evaluated whether the two-port joint attractor resided linearly on the spatial segment directly interpolating the two single-port attractors. Normalized distance ($d/$|seg|) was measured across 5 random seeds.

Joint Port Pairing	**Port Type Class**	**Normalized Segment Distance (d/|seg|)**
$O \downarrow$ with $F \uparrow$	Arithmetic × Local Geometric	0.11
$O \downarrow$ with $K \downarrow$	Arithmetic × Global Geometric	0.29
$p \downarrow$ with $M \downarrow$	Arithmetic × Scale	0.20

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Joint Port Pairing	Port Type Class	Normalized Segment Distance (d/|seg|)
$O\downarrow$ with $p\downarrow$	Arithmetic × Arithmetic (Control)	0.35
$F\uparrow$ with $K\downarrow$	Geometric × Geometric (Control)	0.25

Every pairing landed mathematically on the segment. The largest departure (0.35) belonged to the arithmetic-by-arithmetic control pair, utterly inverting the registered prediction. The compositional algebra hypothesis is. The observable layer operates exclusively as a flat coordinate system. Joint feedback merely forces a weighted Euclidean compromise between singular optima.

9. Structural Algebras: May, Does, and Seen

The physical graph representation of deterministic software relies on transition topologies up to isomorphism. The dictionary between a binary increment counter and a Gray-counter toggle sharing a shape of (8,) was constructed natively as $\sigma(x) = x \oplus (x \gg 1)$, discovering the Gray code from topology without arithmetic. Open Mealy machines extend this classification to input-indexed, output-labeled graphs. Running a search over all 96 triples of simultaneous relabelings between a serial subtractor and a serial adder returns exactly four dictionaries representing the two classical subtraction-via-addition identities (e.g., two's-complement logic). A negative control against a D-latch returned exactly 0 of 96 dictionaries, providing a formal equivalence boundary.

Discrete mechanics partitions reality into three fundamentally interacting algebraic layers: Constraints (MAY), Execution (DOES), and Observation (SEEN).

Constraints define what may happen. They operate as a complete distributive lattice on finite prefix-closed behavioral languages. Verified exhaustively across thousands of triples, the lattice laws—commutativity, associativity, absorption $A \wedge (A \vee B) = A$, distributivity, and modularity—experience strictly zero violations.

The hypothesis that a lattice admits no additive valuation unless it is graded was evaluated. Lattices routinely carry valuations satisfying $v(A) + v(B) = v(A \wedge B) + v(A \vee B)$. This hypothesis is. The correct algebraic prohibition relies solely on idempotence: $f(A) = f(A \wedge A) = 2f(A) \Rightarrow f(A) = 0$. Any additive homomorphism from an idempotent composition law into the additive reals is necessarily trivial. The interface admission price cannot join the accumulating execution ledger.

Execution answers what does happen, forming an accumulating monoid. Execution acts on constraints by precomposition (inverse image). The inverse image $T^{-1}(\cdot)$ acts as a perfect lattice homomorphism, preserving both meet and join with zero violations in 3,000 trials. Forward image $T(\cdot)$ fails to preserve the meet operation in 733 of 3,000 trials, failing exactly at the boundaries where the transition is non-injective (the crushes).

The legality of ahead-of-time software compilation is dictated by this contravariance. Ahead-of-time verification is sound if and only if the admissible set is closed under execution: $T(A) \subseteq A$. Empirical separation verifies 387 closed cases at 100.0% soundness, and 3,613 open cases at 0.0% soundness. Subject reduction is thereby reached as the necessary precondition for compilation.

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The hypothesis that operations partition into a binary of execution verbs and constraint verbs, distinguished by idempotence, was tested. A broader evaluation exposes at least four operation classes, including a pathological parity gate belonging to neither side. The hypothesis is. The defining distinction lies between the compilation operator ν (largest invariant subset) and the orbit closure μ. Compilation ν is an interior operator: it is idempotent, shrinking, monotone, preserves meet exactly, and strictly fails join. Measurement yields $\nu(A \cup B) = \supset \nu(A) \cup \nu(B) =$.

Observation answers what is seen, operating as a quotient space over the executing state. The integration of the pipeline—MAY → DOES → SEEN—is strictly order-dependent. Restricting an admissible state then observing it differs from observing then restricting it in 49.1% of 2,000 trials (e.g., returning $\{0,3\}$ versus $\{0,2,3\}$ for the same variables).

This structural non-commutation of the pipeline constitutes the formal origin of the Entropy Seam. The spectrum survives movement between levels; the toll is level-relative, recording what does not survive. The physical loss is not intrinsic to the microstate of the object; it is created by the morphism. Entropy S_π is therefore defined as the obstruction to a measurement commuting with coarse-graining.

10. Unmeasured Setups and Proposed Operations

The algebraic structures dictate specific proposed experiments whose empirical outcomes are not yet provided. Because these tests remain un-executed, no quantitative conclusion is drawn.

Feedback Composition on Open Machines

Setup: Couple the output of Mealy machine M_1 back to its own input via a serial feedback interface I_f.**Constraints:** Interface I_f must act as a validating, non-total, non-identity contract over the wire message alphabet.**Metric to be measured:** The count of distinct closed state trajectories and the resulting periodic cycle spectrum under restricted composition.**Conclusion:**

Disjunctive Checkability in Static Analyzers

Setup: Run a static analysis over a program containing disjunctive loop invariants $A \vee B$ over non-prefix-closed sets.**Constraints:** Invariants A and B must individually fail to satisfy the closed-form preservation condition $T(X) \subseteq X$, while the joint space maintains cyclic coverage.**Metric to be measured:** Pass/fail validation rate of the joint invariant $A \vee B$ versus the sum of individual invariant validations.**Conclusion:**

Hardware Realization of Idempotent Constraint Instructions

Setup: Profile instruction execution times on real processor architectures executing repeated invariant check routines.**Constraints:** Invariant checks must act purely as a projection onto a fixed set (ASSERT style) satisfying $F \circ F = F$.**Metric to be measured:** The energy cost and cumulative cycle count of executing the projection constraint twice sequentially versus executing it once.**Conclusion:**

The Dynamical Equivalence of Ledger Depth and Time

Setup: Trace an executing de Bruijn sequence program and record its cumulative computational bit expansion rate in memory.**Constraints:** The program must execute under a constant, non-zero continuous trace loss rate $\mathrm{tr}(M) = -0.165515$.**Metric to be measured:** The linear correlation scaling coefficient between discrete execution steps and the geometric address space volume expansion.**Conclusion:**

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